

Adaptive Night Light

Measure, decide, illuminate, and explain why

The circuit makes its internal decision visible without requiring Serial.

Lesson	009 — integration project	Time	150–210 minutes
Prerequisites	Lessons 001–008	Board	Arduino Mega 2560
Evidence	voltage, raw sample, mode, PWM	Status	host verified; bench open
Safety	E1: USB-powered passive sensor and current-limited LEDs only		

1 Mission and evidence chain

Build a night light whose behavior can be explained at every layer:

ambient light → A0 voltage → raw sample → filtered permille → hysteresis mode → D9 PWM.

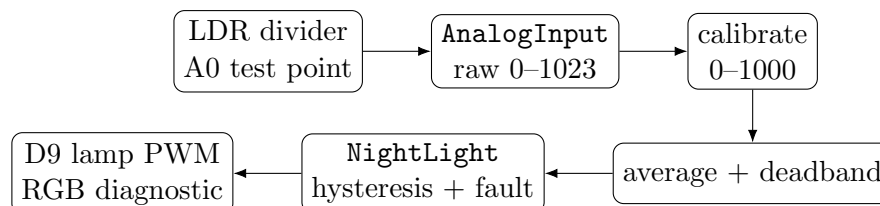
The white lamp is the primary output. A common-cathode RGB LED is the diagnostic output: blue means **Ready**, green means **Active**, and red means **SensorFault**. The named sensor test point is the divider junction at A0; the named safe-state test point is D9 relative to GND.

By the end, you can:

- compose `AnalogInput`, calibration, moving average, deadband, `NightLight`, `PwmOutput`, and `RgbLed`;
- explain why calibration, filtering, deadband, and hysteresis solve different problems;
- reproduce a decision from an ordered sample trace;
- distinguish sensor open/short evidence from ordinary darkness or light;
- prove resource acquisition separately from the lamp's safe state;
- build, upload, monitor, and capture optional logs entirely from the CLI.

Safety checkpoint. Disconnect USB and all other power before wiring. Use one 330Ω resistor for the white LED and one for each RGB die. The sensor divider must remain between the Mega's 5 V and GND rails; never apply an external signal to A0. Stop for heat, odor, resets, flicker unrelated to the lesson, or any measured voltage outside 0–5 V.

2 System composition



The canonical sketch deliberately reads like this diagram. Its `loop()` first calls `observeLight()`, then `decideLighting()`, then `actuateLighting()`. Hardware ownership remains in the endpoints; `NightLight` receives a value and knows nothing about Arduino pins.

2.1 Why four transformations?

Stage	Question answered	What it does not prove
Calibration	What normalized light does this raw value represent?	Stability over time
Moving average	What recent level does a noisy stream support?	A mode decision
Deadband	Is this tiny change worth reporting?	Lamp on/off policy
Hysteresis	Has light crossed the appropriate boundary for the current mode?	Electrical output

3 Authoritative pencil build plate

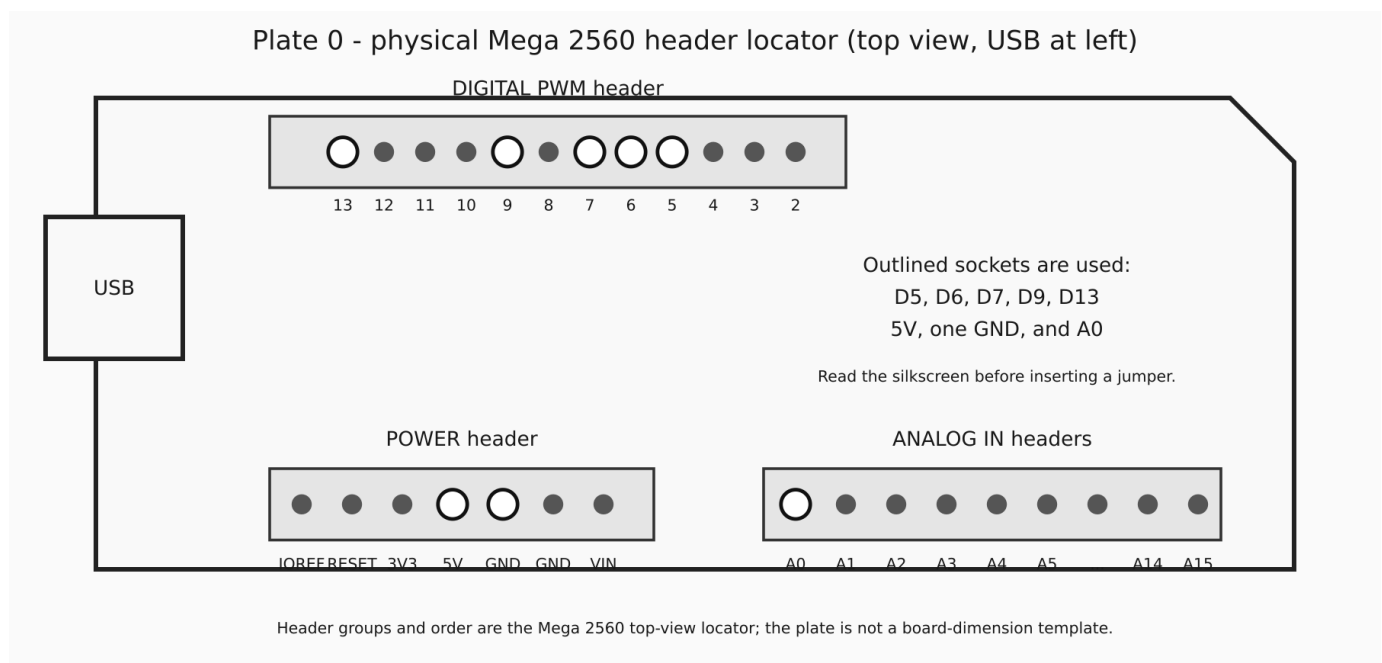


Figure 1: Plate 0 locates the used sockets in their physical Mega 2560 header groups and order. Orient the board with USB at left and still confirm the printed silkscreen before inserting a jumper.

4 Parts and literal wiring

Quantity	Part	Requirement
1	Mega 2560	USB powered
1	photoresistor	identified passive LDR; record its datasheet or kit reference
1	resistor	10 k Ω , divider lower leg
1	white LED	identified polarity and forward-voltage range
1	common-cathode RGB LED	verify package pinout; lead order is not universal
4	resistors	330 Ω , one per LED die
1	multimeter	DC voltage and unpowered resistance/continuity

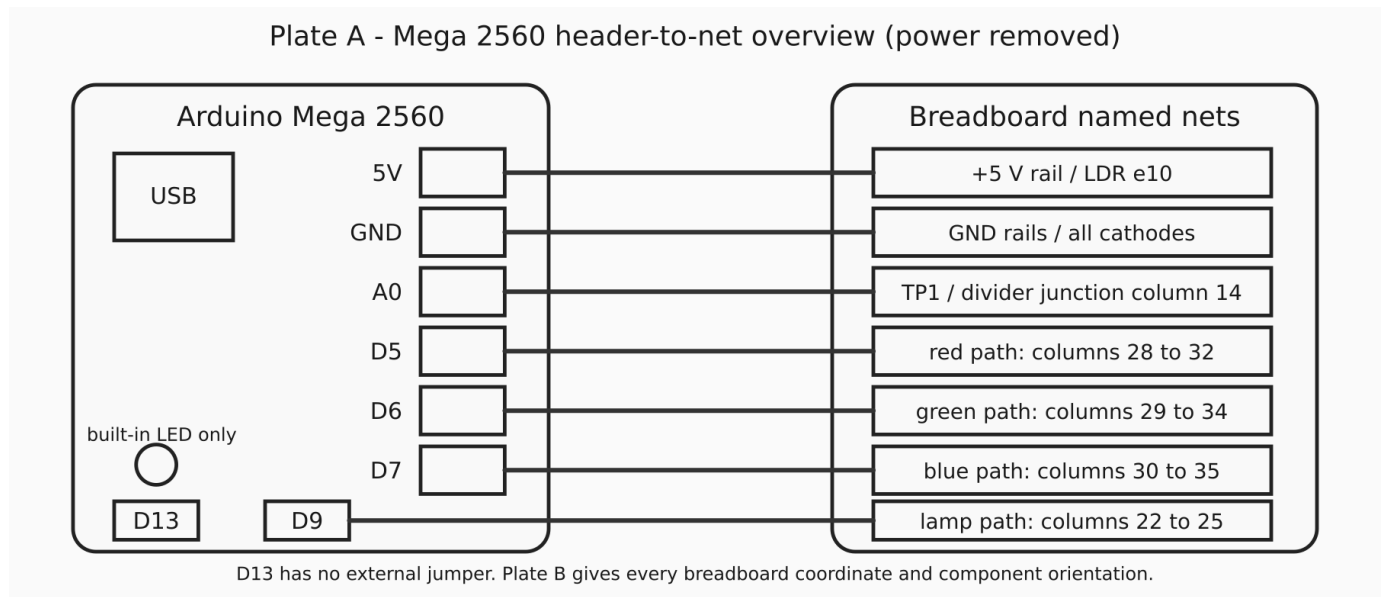


Figure 2: Plate A separates every used physical Mega header from its named breadboard net. D13 is the built-in LED and has no external jumper.

Specimen gate. The authorized Elegoo family listing proves that an LDR, white LED, and RGB LED belong to the kit; it does not prove the selected specimen's electrical identity. Before wiring, record the LDR package/photo and measured dark/light resistance; white LED package, anode/cathode, and V_f ; and RGB package/photo, viewing orientation, individual lead numbers, common-cathode lead, each color lead, and each color's V_f . The build plate uses the verified left-to-right RGB order R–K–G–B at e32–e35, viewed from the flat/package reference recorded on the card. If the specimen differs, stop: revise both the build plate and manifest before applying USB power.

Coordinate net manifest. The breadboard's a–e holes in one numbered column are connected; f–j form a separate group across the center trench. Power-rail breaks vary by board. Test continuity unpowered and add only the rail bridge required for the used segment.

Net	Exact build-plate path	Cross-check
rails	Mega 5 V to used top + rail; Mega GND to used top – rail; join used bottom – rail to the same GND	continuity to named Mega header; no 5 V–GND continuity
sensor	+ rail to LDR e10; other LDR lead e14; A0 to a14/TP1; 10 k Ω from b14 to – rail	e14/a14/b14 are one junction; TP1 is measured only to GND
lamp	D9 to a22; 330 Ω e22–e25; white anode a25; white cathode f25 to bottom – rail	LED straddles trench; TP-LAMP is a25 to GND
RGB red	D5 to a28; 330 Ω e28–e32; verified red lead e32	resistor measured unpowered
RGB return	verified common cathode e33 to bottom – rail	continuity to Mega GND
RGB green	D6 to b29; 330 Ω d29–d34; verified green lead e34	resistor measured unpowered
RGB blue	D7 to c30; 330 Ω c30–c35; verified blue lead e35	resistor measured unpowered
D13 evidence	Mega built-in LED only; no external breadboard jumper	250 ms pulse, then 750 ms dark separator

The fixed 10 k Ω lower leg limits divider current to at most 0.5 mA at 5 V even if the LDR is shorted. Its

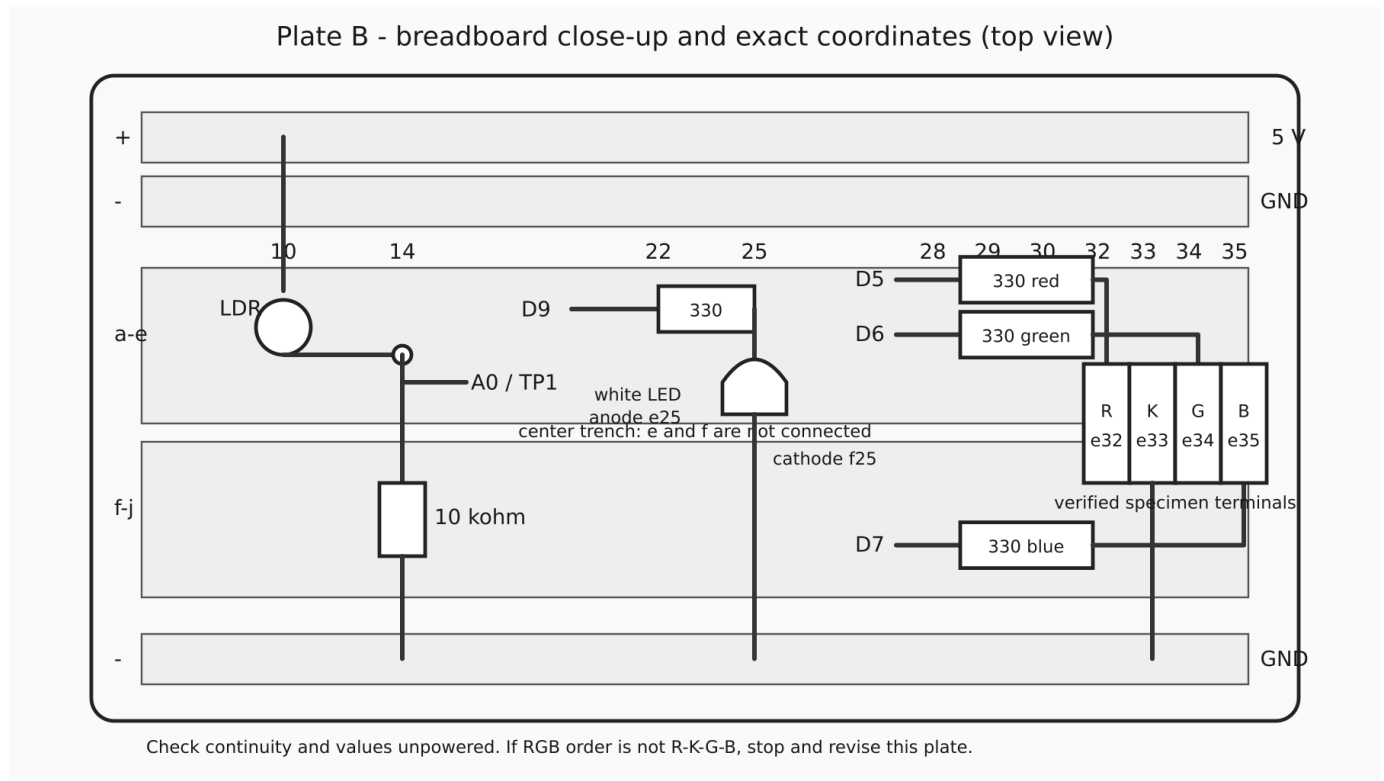
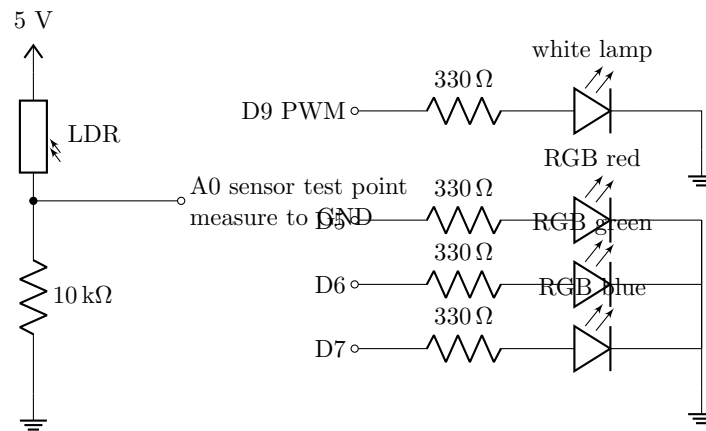


Figure 3: Plate B enlarges breadboard topology, coordinates, polarity, and resistor placement. Cross-check both plates against the manifest below.

Thevenin source impedance is never above 10k Ω , matching the conservative ADC source-impedance guidance used in this lesson.



More light lowers LDR resistance, raises A0 voltage, and raises the normalized `lightPer mille` value. If your identified sensor behaves differently, change the divider orientation and calibration together; do not silently invert only the software.

4.1 Current and timer budget

For each LED die, calculate $I = (V_{OH} - V_f)/330\Omega$ with the actual part's minimum V_f and a conservative loaded V_{OH} . Record the worst simultaneous case: the lamp plus up to three RGB dies. Keep margin below every ATmega2560 pin, port-group, and package operating limit; absolute maximum ratings are not design

targets. This lesson additionally requires the calculated and measured current to remain at or below 15 mA per channel.

Channel	measured V_f	calculated worst current	measured active current
white / D9			
red / D5			
green / D6			
blue / D7			

Worst simultaneous case named: _____. Calculated total: _____ mA. Measured total: _____ mA. Record the actual combination observed; do not add four single-channel measurements if the firmware never commands that combination.

D9 uses the Mega's Timer2 PWM resource. D5, D6, and D7 use other PWM timers. Do not add `PiezoSounder` without resolving its exclusive Timer2 claim: the conflict must fail during initialization, before an active waveform.

5 Decision contract

```

adk::NightLightConfig nightLightConfig;
nightLightConfig.turnOnBelowPer mille = 350;
nightLightConfig.turnOffAbovePer mille = 450;
nightLightConfig.minimumDuty          = 24;
nightLightConfig.maximumDuty          = 192;

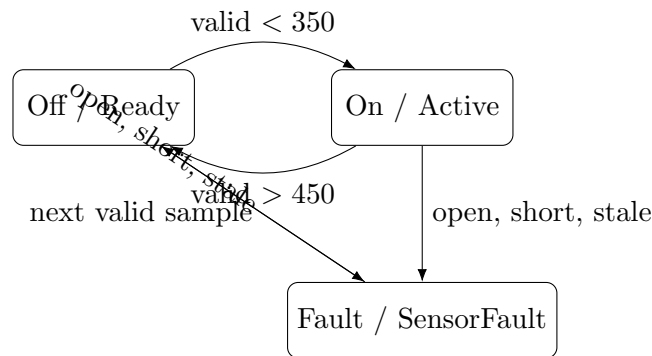
adk::NightLight nightLight (nightLightConfig);

const adk::NightLightInput observation (
    filteredLight,
    adk::LightSampleState::Valid);

nightLight.update (observation);
const adk::NightLightSnapshot decision = nightLight.snapshot ();

```

NightLight starts **Off**. After successful acquisition the built-in D13 LED pulses for 250 ms and stays dark for a 750 ms separator; normal behavior never drives it. A valid value strictly below 350 permille changes it to **On**; it remains on through the middle band and changes off only strictly above 450. Equality does not cross either boundary. This memory is hysteresis, not filtering.



Any explicit **BelowRange**, **AboveRange**, or **Stale** sample enters **Fault**, forces duty zero, and selects the red diagnostic. The next valid sample first recovers to **Off**; darkness must then satisfy the ordinary turn-on rule. This prevents a recovered sensor from inheriting an old active command.

6 Deterministic trace exercise

Use an eight-sample moving average and an eight-permille deadband. For this short worksheet, the “filtered” column is supplied so the state-machine reasoning remains visible.

Step	raw	filtered	state	prior mode	next mode	duty
1	700	680	Valid	Off	_____	_____
2	360	440	Valid	Off	_____	_____
3	240	340	Valid	Off	_____	_____
4	410	400	Valid	On	_____	_____
5	520	451	Valid	On	_____	_____
6	1023	—	AboveRange	Off	_____	_____
7	220	330	Valid	Fault	_____	_____

Replay the same ordered samples twice in the host test. Equal configuration and samples must produce equal snapshots. No test reads wall time, sleeps, or depends on Serial delivery.

7 Fault matrix

Fault or boundary	Expected decision	Lamp	RGB
A0 near 0	suspected divider fault; not ordinary darkness	duty 0	red
A0 near 1023	suspected divider fault; not ordinary daylight	duty 0	red
missing fresh value	Stale	duty 0	red
350 exactly while off	strict boundary does not turn on	duty 0	blue
450 exactly while on	strict boundary does not turn off	active	green
D9 already claimed	initialization fails before operation	high impedance	unavailable or inactive
Timer2 exclusively claimed	initialization fails before PWM	high impedance	inactive
fixed 120 s deadline	outputs inactive for 250 ms, then reverse shutdown	high impedance	high impedance
reset or shutdown	reset repeats acquisition; shutdown releases ownership	high impedance	high impedance

Do not overclaim the RGB evidence. Blue proves that all objects initialized and the controller selected **Ready**. It does not prove D9 is electrically safe. Measure D9 to GND separately: zero duty while initialized should be near 0 V. For high impedance, remove USB, install the one-at-a-time 10 k Ω /10 k Ω midpoint fixture described below, inspect it, restore USB, and measure the midpoint. Never use an ohmmeter on a powered circuit.

8 CLI-only workflow

No IDE is required. From the repository root:

```
make bootstrap
make host-test
make arduino-Lesson009AdaptiveNightLight
arduino-cli compile --fqbn arduino:avr:mega --library . \
  --build-property \
    compiler.cpp.extra_flags=-DADK_LESSON009_ACCEPTANCE_TELEMETRY=1 \
    examples/Lesson009AdaptiveNightLight
make upload EXAMPLE=Lesson009AdaptiveNightLight PORT=/dev/ttyACM0
make monitor-describe PORT=/dev/ttyACM0
make monitor PORT=/dev/ttyACM0 BAUD=115200
make serial-log PORT=/dev/ttyACM0 BAUD=115200 \
  SERIAL_LOG=build/serial/lesson009.log
make lessons
make site
```

The circuit supplies the acceptance evidence even if **monitor** is never run. The compile-time acceptance build emits only **raw=** ADC counts, bounded to one record per 250 ms and by the fixed 120 s deadline. It is supporting evidence for the TP1 voltage observation, never the acquisition or safe-state proof. The default build has telemetry disabled.

9 Bench procedure: predict, observe, interpret

9.1 Stage A — inspect without power

1. ☐ Confirm the exact Mega pins, every 330Ω resistor, recorded RGB R–K–G–B specimen orientation, LDR divider orientation, and every build-plate coordinate.
2. ☐ Measure the divider's lower resistor: expected _____, observed _____.
3. ☐ Confirm no continuity from 5 V directly to GND and no LED die lacks its series resistor.
4. ☐ Cross-check each manifest net with a second person or a second pass; mark every row only after continuity agrees.
5. ☐ Predict the A0 voltage in bright and covered conditions.

9.2 Stage B — acquisition evidence

Power from USB. Do not upload until the unpowered checks pass.

1. ☐ Build and upload with the CLI targets above.
2. ☐ Predict: D13 lights for 250 ms, remains dark for the 750 ms separator, then valid moderate light selects blue.
3. ☐ Observe D13 timing: on _____ ms; separator _____ ms.
4. ☐ Observe RGB: _____.
5. ☐ Interpret: _____.

9.3 Stage C — sensor and decision evidence

1. ☐ Measure A0 to GND under bright light: predicted _____ V / _____ counts; observed _____ V / _____ counts. Cover the LDR: predicted _____ V / _____ counts; observed _____ V / _____ counts. Interpret whether count direction and measured rail ratio agree: _____.
2. ☐ Move light slowly through the on boundary. Record A0 voltage when the lamp becomes active: _____ V.
3. ☐ Reverse direction. Record voltage when it becomes ready: _____ V. Explain the different boundaries: _____.
4. ☐ Observe green with the lamp waveform. At D9, record PWM high level _____ V; predicted duty _____%; observed duty _____%. Interpret: _____.

9.4 Stage D — fault and safe-state evidence

Remove USB before changing the sensor connection. Simulate only one fault at a time, then restore wiring unpowered.

1. ☐ With USB removed, move only the +-rail jumper from LDR e10 to an unused insulated row, leaving e14/a14/b14 and the $10\text{ k}\Omega$ lower leg intact. Restore USB. Predict red and lamp duty zero. Observe: _____.
2. ☐ Remove USB. Restore e10, then place one jumper e10–e14 across only the LDR terminals. Inspect, restore USB, and predict red with duty zero. Observe: _____.
3. ☐ Restore normal wiring. Confirm recovery goes through blue before darkness can select green.
4. ☐ Record D9 initialized at zero duty: _____ V. Let the fixed 120 s deadline expire. Observe lamp and RGB inactive for at least 250 ms before release: _____.
5. ☐ Remove USB. Disconnect the white LED/resistor path from D9. Install two $10\text{ k}\Omega$ resistors in series from 5 V to GND and connect their midpoint to D9 only. Inspect, restore USB, and after the fixed deadline

record midpoint voltage _____ V. A value near half the measured rail supports high impedance; near a rail does not. Remove USB before removing the fixture.

6. ☐ Repeat the midpoint test one output at a time for D5, D6, and D7, never with an LED path attached. Record: D5 _____ V; D6 _____ V; D7 _____ V.
7. ☐ Press reset after restoring the normal circuit unpowered. Record the repeated D13 pulse/separator and Ready recovery: _____.

This procedure detects the named open and short simulations because the fixed divider leg drives A0 toward a rail. A broken A0 jumper can float at a plausible value and is not universally detectable with one ADC channel. Even then, every accepted sample remains bounded and output duty cannot exceed the configured maximum; inspect continuity unpowered when evidence disagrees.

10 Diagnosis by layer

Observation	Most useful next check	Reason
no RGB at startup	resource conflict, polarity, resistor, common cathode	acquisition evidence is absent
red RGB, lamp off	A0 voltage and divider continuity	controller reports sensor evidence invalid
blue/green changes, lamp dark	D9 waveform, LED polarity, lamp resistor	decision exists; actuator path is suspect
lamp chatters near threshold	raw range, average window, dead-band, threshold order	separate noise from policy
A0 changes, mode does not	calibration direction and normalized values	electrical sensing works; mapping may not
unexpected reset	disconnect power and inspect shorts/current budget	software evidence cannot clear an electrical fault

11 Design exercises

1. Plot measured A0 voltage against normalized permille at five light levels. Which calibration points should be learned from your room?
2. Change only the moving-average window. Predict latency and noise before compiling; then record both.
3. Change only the hysteresis gap. Find the smallest gap that avoids chatter in a repeatable hand-shadow trace.
4. Propose a stale-sample policy using explicit time. State the maximum valid age and add rollover cases before changing the interface.
5. Add a host trace for one sensor fault during active PWM. Prove the first fault snapshot commands zero duty and the first recovery snapshot is `Off`.

12 Acceptance record

Field	Record	Field	Record
Board/revision		Date	
USB source/cable and measured 5 V rail		Reviewer	
Arduino CLI/core		ADK revision	
Multimeter make/model/asset ID		LDR package/photo/resistance	
Acceptance sketch artifact/hash		Logic probe/scope make/model/asset ID	
White LED package/polarity/ V_f		RGB package/photo/R-K-G-B pin map	
Flash bytes		Static RAM bytes	

- ☐ Unpowered wiring and current calculations passed.
- ☐ Ready, Active, and SensorFault indications matched the table.
- ☐ A0 voltage supported the raw-sample interpretation.
- ☐ Hysteresis produced distinct on and off boundaries.
- ☐ Open and short sensor tests forced lamp duty zero.
- ☐ Resource acquisition and D9 safe state were proven separately.
- ☐ Reset, shutdown, and power removal observations were recorded.
- ☐ Measured rail, TP1 light extremes, each $330\ \Omega$ resistor, each active-channel current, and worst simultaneous current total were recorded.

Observed deviations and unresolved checks:

Release status. Compilation, host replay, and a generated PDF do not constitute physical verification. Until a person completes this card on the named Mega 2560 and records measurements, this lesson remains host verified with hardware acceptance open.

13 Use with

Use the maintained HTML lesson for searchable API links, errata, source downloads, and exact CLI targets. Use this PDF at the bench for predictions, schematics, worksheets, and recorded evidence. The canonical editable sketch is `examples/Lesson009AdaptiveNightLight/` rather than a duplicate in this document.

- Lesson 009 HTML
- Canonical sketch
- NightLight API
- Arduino Mega 2560 datasheet
- ATmega2560 primary product page